

CR-001: Client-Side Search and Filtering of the Admin Student List

1. Change Request Details

Field	Details
Change Request ID	CR-001
Title	Client-Side Search and Filtering of the Admin Student List
Priority	High
Difficulty	Low
Change Type	Frontend Enhancement
Affected Version	Student Management System v1.0 baseline
Dependency	Existing GET /api/students/ endpoint

2. Business Requirement

Administrators currently need to manually scan the complete student table to locate a particular student. As the number of students increases, this becomes time-consuming and may increase the possibility of selecting the wrong student.

The system should provide an easy way for administrators to quickly locate students using registration number, name, or email address.

3. Current Behaviour

- `DashboardPage.jsx` loads students using GET /api/students/.
- The complete student list is displayed in a table.
- No search or filtering facility is currently available.
- Administrators must manually scan the table.
- Existing Edit and Delete operations use `student_id`.

4. Requested Change

Add a client-side search facility above the administrator's student table.

The implementation must:

- Search by registration number, first name, last name, full name, and email.
- Support partial and case-insensitive matching.
- Ignore leading/trailing spaces and normalize repeated spaces.
- Filter already-loaded records without an API request for every keystroke.
- Display the number of matching records.
- Provide a Clear Search function.
- Display a clear no-result message.
- Preserve existing Edit and Delete functionality.
- Ensure Edit/Delete continue to use the correct `student_id`.
- Maintain existing administrator authentication and authorization.

5. Scope

In Scope

- Dashboard search input.
- Client-side filtering.
- Partial and case-insensitive matching.
- Name, registration number, and email searching.
- Result count.
- Clear Search.
- No-result state.
- Responsive search interface.
- Preservation of existing CRUD functionality.

Out of Scope

- Database changes.
- Backend changes.
- New API endpoints.
- Server-side searching.
- Pagination.
- Server-side sorting.
- Changes to authentication or authorization.

6. Expected / Potentially Affected Files

- `frontend/src/pages/DashboardPage.jsx`
- `frontend/src/components/StudentSearchBar.jsx` (*optional*)
- Existing dashboard stylesheet

Students may modify additional files if required. Any additional files must be identified and justified in the Implementation Report.

7. Functional & Security Requirements

- **FR-001:** Search input shall be available above the student table.
- **FR-002:** Search shall support registration number, first name, last name, full name, and email.
- **FR-003:** Search shall be case-insensitive and support partial matching.
- **FR-004:** Leading/trailing and repeated spaces shall not incorrectly prevent matching.
- **FR-005:** Filtering shall occur against already-loaded records.
- **FR-006:** Clearing the search shall restore the complete list.
- **FR-007:** The interface shall display matching record count and a no-result message.
- **FR-008:** Edit and Delete shall continue to use the correct `student_id`.
- **SEC-001:** Existing administrator authentication and authorization shall remain unchanged.
- **SEC-002:** The feature shall not expose additional sensitive information or use unsafe HTML rendering.

8. Acceptance Criteria

The CR is successfully implemented when:

- **AC-001:** Administrators can search by registration number, name, and email.
- **AC-002:** Search is case-insensitive and tolerates unnecessary spaces.
- **AC-003:** Partial matching returns the appropriate records.
- **AC-004:** Result count matches the visible records.
- **AC-005:** No-result searches display a clear message and Clear Search option.
- **AC-006:** Clearing the search restores the complete list.
- **AC-007:** Edit and Delete continue to affect the correct student.
- **AC-008:** Existing administrator CRUD and authorization continue to work.

9. Expected Outcome

Administrators can quickly locate students without manually scanning the complete table, while existing API, database, authentication, authorization, and CRUD functionality remain unchanged.

10. Implementation Report Requirement

After implementation, the Final Exam Software Change Implementation Report shall include:

- Implementation summary.
- Affected files.
- Important technical/security changes.
- Test cases and results.
- Acceptance criteria verification.
- Relevant screenshots/evidence.
- Final status and outstanding issues.